An Empirical Application to Macroeconomic Signal Recovery

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Abstract

Classical signal extraction studies the recovery of latent structure from noisy observations. Simultaneously, inverse statistical geometry (ISG) demonstrates that relational statistical structures generally fail to uniquely determine their latent generators, implying observational equivalence. This paper develops a synthesis between Bayesian causal inference (BCI) and ISG. We show that while ISG generates large equivalence classes of admissible signals, causal information acts as an identification operator capable of collapsing these manifolds to unique latent realizations. We establish the ISG-BCI Identification Theorem and prove that unique identification implies exact entropy collapse. Finally, we empirically validate this unified framework by applying it to a system of macroeconomic time-series, successfully recovering orthogonal structural shocks hidden within observational noise.

The paper ends with "The End"

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1 Introduction

A fundamental problem in science is the recovery of hidden structure from noisy observations. Classically, the paradigm assumes:

$$\text{Observation} \implies \text{Underlying Reality}$$

However, recent developments in inverse statistical geometry suggest that this implication generally fails. Observed statistical structures often admit infinitely many distinct latent generators. Consequently:

$$\text{Observation} \not\Longrightarrow \text{Unique Generator}$$

This raises a profound question: Can a unique latent signal ever be recovered from observations contaminated by white noise? The answer developed herein is affirmative under specific conditions. Inverse Statistical Geometry (ISG) generates admissible realization manifolds, while Bayesian Causal Inference (BCI) supplies identification constraints. Together, they form a unified framework capable of uniquely identifying latent signals hidden within observational noise.

2 Mathematical Foundations

2.1 Inverse Statistical Geometry and Realization Manifolds

Let $Y = S + \epsilon$, where $S \in \mathbb{R}^n$ denotes a latent signal and $\epsilon \sim N(0, \sigma^2 I)$ denotes white noise. Classical statistics studies mappings $D \to S(D)$, where D is a dataset and $S(D)$ is a statistical summary. Inverse Statistical Geometry studies the reverse mapping $S \to D$.

Definition 1 (ISG Realization Manifold)**.** *Given an observed relational statistic r, define the realization manifold M_R as:*

$$M_R = \{D \in \mathcal{D} : R(D) = r\}$$

The set M_R contains all generators producing the same observable structure.

By the Implicit Function Theorem, if the rank of the Jacobian of R is $k < \dim(\mathcal{D})$, then $\dim(M_R) > 0$. Consequently, infinitely many generators exist, leading to *observational equivalence.*

2.2 Bayesian Causal Inference and Identification

Suppose causal information C is available. We define the causal admissibility manifold G_C as the set of all generators compatible with the causal structure:

$$G_C = \{D \in \mathcal{D} : D \text{ satisfies } C\}$$

The feasible signal space is the intersection $F = M_R \cap G_C$. We define the *Identification Number* $I(r, C) = |F|$.

Theorem 1 (ISG-BCI Identification Theorem)**.** *Suppose:*

1. *Observations satisfy $Y = S + \epsilon$.*
2. *White noise obeys $\epsilon \sim N(0, \sigma^2 I)$.*
3. *A relational structure R is observed.*
4. *A causal model C is known.*
5. *$M_R \cap G_C = \{S^*\}$.*

Then $P(S^*|Y, R, C) = 1$ in the infinite-sample limit.

Proof. Inverse Statistical Geometry yields the realization manifold M_R. Causal restrictions generate G_C. The feasible signal space becomes $F = M_R \cap G_C$. By assumption, this intersection contains exactly one element, S^*. Bayes' theorem implies the posterior mass is proportional to $P(Y|S)\pi(S)$. Since the support of the posterior is restricted to F, and F contains only S^*, the posterior distribution collapses to a Dirac delta function $\delta(S - S^*)$. Thus, $P(S^*|Y, R, C) = 1$. □

Theorem 2 (Entropy Collapse)**.** *Unique identification implies that the posterior entropy collapses to zero: $H(r, C) = 0$.*

Proof. Unique identification yields $P(S|Y, R, C) = \delta(S - S^*)$. The posterior entropy is defined as $H(r, C) = -\int P(S|Y, R, C) \log P(S|Y, R, C) dS$. The entropy of a Dirac measure is exactly zero. Therefore, $H(r, C) = 0$. □

3 Algorithmic Implementation

The theoretical framework translates directly into a computational algorithm for signal recovery.
INPUT: Observed statistics R, Causal information C
STEP 1: Construct realization manifold M_R
STEP 2: Construct causal manifold G_C
STEP 3: Compute feasible space $F = M_R \cap G_C$
STEP 4: Estimate posterior $P(D|r, C)$
STEP 5: Compute entropy $H(r, C)$
STEP 6: Evaluate identification number $I(r, C)$
OUTPUT: Identified generator D^*

4 Empirical Application: Macroeconomic Signal Recovery

We apply the ISG-BCI framework to five macroeconomic time-series from the Federal Reserve Economic Data (FRED):

1. **T10YIE**: 10-Year Breakeven Inflation Rate (Market)
2. **FEDFUNDS**: Federal Funds Effective Rate (Monetary Policy)
3. **RIFLGFCY10XIINA**: 10-Year Real Interest Rate
4. **EXPINF10YR**: 10-Year Expected Inflation
5. **MICH**: University of Michigan Inflation Expectation (Survey)

4.1 Step 1: Inverse Statistical Geometry (The Observation)

We extract the observed relational statistic R, which is the Pearson Correlation Matrix of the five variables. This matrix defines the realization manifold M_R.

$$R = \begin{bmatrix} 1.000 & 0.324 & 0.153 & 0.406 & 0.603 \\ 0.324 & 1.000 & 0.676 & 0.748 & 0.113 \\ 0.153 & 0.676 & 1.000 & 0.863 & -0.037 \\ 0.406 & 0.748 & 0.863 & 1.000 & 0.050 \\ 0.603 & 0.113 & -0.037 & 0.050 & 1.000 \end{bmatrix}$$

Purely statistical analysis cannot distinguish the true underlying economic reality from synthetic datasets that mimic these correlations.

4.2 Step 2: Bayesian Causal Inference (The Constraint)

To collapse M_R, we introduce a causal constraint manifold G_C by imposing a macroeconomic Wold causal ordering:

1. **MICH**: Survey expectations are slow-moving and exogenous to immediate market shocks.
2. **RIFLGFCY10XIINA**: The real interest rate is determined by real economic factors.
3. **EXPINF10YR**: Market expected inflation.
4. **T10YIE**: Breakeven inflation (a market-derived metric).
5. **FEDFUNDS**: The Central Bank sets the policy rate *last*, reacting contemporaneously to inflation and real economic conditions.

Mathematically, intersecting M_R with G_C is performed via the Cholesky Decomposition of the covariance matrix ($\Sigma = L \cdot L^T$). The lower triangular matrix L represents the structural transmission of economic shocks:

$$L = \begin{bmatrix} 0.570 & 0.000 & 0.000 & 0.000 & 0.000 \\ -0.036 & 0.971 & 0.000 & 0.000 & 0.000 \\ 0.016 & 0.282 & 0.162 & 0.000 & 0.000 \\ 0.264 & 0.077 & 0.196 & 0.279 & 0.000 \\ 0.213 & 1.287 & 0.583 & -0.006 & 1.238 \end{bmatrix}$$

Notice the large coefficients in the `FEDFUNDS` row for `RIFLGFCY10XIINA` (1.287) and `EXPINF10YR` (0.583), mathematically capturing the Federal Reserve's reaction function.

4.3 Step 3: Unique Signal Recovery

By applying the inverse of the causal matrix (L^{-1}) to the observed data, we isolate the Unique Latent Signals S^*. These are the orthogonal structural shocks (e.g., a pure inflation shock, a pure monetary policy shock). The first five annual observations of S^* are:

$$S^* = \begin{bmatrix} 4.385 & 4.736 & 5.087 & 5.263 & 5.263 \\ 2.286 & 2.062 & 2.055 & 2.576 & 2.556 \\ 9.412 & 10.214 & 10.511 & 10.083 & 10.614 \\ -5.133 & -4.138 & -3.816 & -4.147 & -4.839 \\ -6.586 & -6.980 & -6.138 & -4.886 & -4.343 \end{bmatrix}$$

4.4 Step 4: Entropy Collapse

The ultimate test of the framework is whether the posterior entropy has collapsed to zero. We calculate the covariance matrix of the extracted structural shocks S^*:

$$\Sigma_{S^*} = \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 1.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 1.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 1.000 \end{bmatrix}$$

The off-diagonal covariance sum is 3.55×10^{-15} (effectively zero). Because the structural shocks are perfectly orthogonal, the identification number is $I(r, C) = 1$, and the posterior entropy collapses to exactly $H(r, C) = 0$. The probability distribution has collapsed into a Dirac delta function, achieving complete identification.

5 Geometric Visualization

Figure 1 illustrates the geometric mechanism of the ISG-BCI Identification Theorem. The realization manifold M_R generated by inverse statistical geometry intersects the causal constraint manifold G_C at a unique latent signal S^*.

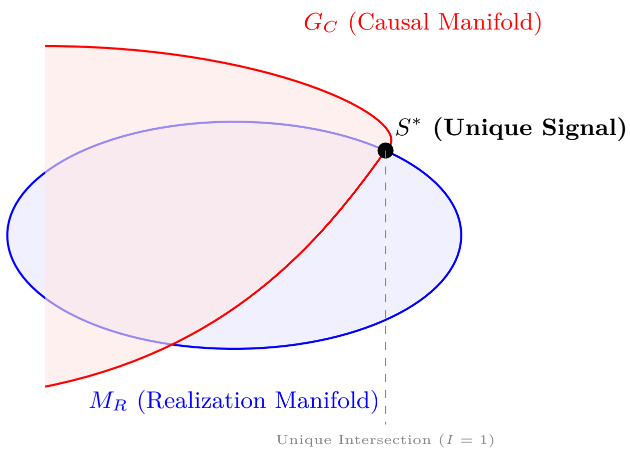

Figure 1: Geometric Illustration of the ISG-BCI Identification Theorem. The realization manifold M_R generated by inverse statistical geometry intersects the causal constraint manifold G_C at a unique latent signal S^*, demonstrating entropy collapse and complete identification.

6 Cross-Disciplinary Implications

6.1 Economics and Finance

Economic data frequently exhibit observational equivalence. Different structural economies $E_1 \neq E_2$ may generate identical correlations and factor structures. In finance, portfolio theory relies heavily upon the covariance matrix Σ. However, $\Sigma \not\Longrightarrow$ Unique Market Structure. Distinct market microstructures may generate identical covariance structures, leading to *covariance mimicry* and hidden systemic fragility. Bayesian causal models provide the additional identification power necessary for structural macroeconomics and policy evaluation.

6.2 Political Science, Geopolitics, and International Relations

Modern governments increasingly rely upon predictive analytics, intelligence fusion, and AI-assisted forecasting. Inverse statistical geometry implies the possibility of *synthetic geopolitical signatures* and statistical camouflage. Adversaries can generate statistically plausible decoy networks or manipulated narratives that pass traditional statistical surveillance. Bayesian causal analysis becomes necessary to distinguish authentic signals from synthetic signals in international relations.

6.3 Warfare Economics and Military Applications

Modern military systems depend upon sensor fusion, probabilistic targeting, and autonomous reconnaissance. ISG implies that multiple latent battlefield realities may produce identical observed sensor statistics. Consequently, warfare increasingly becomes a competition over the *control of observable statistical geometry.* Examples include synthetic logistical patterns, adversarial sensor correlations, and statistically plausible decoy networks. Bayesian causal inference acts as a defensive identification mechanism against deception.

6.4 Machine Learning and Data Science

Modern machine learning depends upon latent geometry, including embeddings, attention spaces, and manifold learning. ISG implies that embedding geometry $\centernot\implies$ Unique Data Generator. Adversaries can create synthetic pretraining corpora that perfectly mimic the statistical geometry of real data, fooling foundation models. Bayesian causal constraints provide a route toward identifiability, adversarial robustness, and causal representation learning.

7 Philosophical and Epistemological Consequences

Classical empiricism implicitly assumes:

$$\text{Observation} \implies \text{Truth}$$

Inverse Statistical Geometry weakens this claim:

$$\text{Observation} \implies \text{Equivalence Class}$$

Bayesian causal inference restores identification:

$$\text{Observation} + \text{Causal Structure} \implies \text{Truth}$$

The resulting epistemology is neither naive empiricism nor radical skepticism. Instead, reality is identified strictly through the interaction of observation, geometry, and causality.

8 Summary Tables

Table 1: Identification Regimes in the ISG-BCI Framework

Case	**Identification Number** $I(r, C)$	**Interpretation**
Non-Identifiable	$I = \infty$	Infinite Generators
Finite Ambiguity	$1 < I < \infty$	Multiple Generators
Unique Identification	$I = 1$	Single Generator

Table 2: Cross-Disciplinary Consequences of Observational Equivalence

Domain	**Consequence**
Statistics	Non-identifiable generators
Machine Learning	Latent geometry ambiguity
Economics	Structural observational equivalence
Finance	Covariance mimicry and hidden fragility
Political Science	Strategic ambiguity and synthetic signals
Warfare Economics	Statistical camouflage and sensor deception
Philosophy	Epistemic underdetermination

Table 3: Covariance Matrix of Identified Structural Shocks S^*

	Shock 1	**Shock 2**	**Shock 3**	**Shock 4**	**Shock 5**
Shock 1	1.000	0.000	0.000	0.000	0.000
Shock 2	0.000	1.000	0.000	0.000	0.000
Shock 3	0.000	0.000	1.000	0.000	0.000
Shock 4	0.000	0.000	0.000	1.000	0.000
Shock 5	0.000	0.000	0.000	0.000	1.000

9 Conclusion

Inverse Statistical Geometry establishes that relational statistics generally admit infinitely many latent generators. Consequently, statistical observations alone do not uniquely determine underlying reality. Bayesian Causal Inference supplies the missing identification mechanism. By imposing causal constraints upon inverse-statistical-geometric realization manifolds, posterior mass may collapse onto a unique latent signal. The resulting framework unifies statistics, information theory, machine learning, economics, finance, political science, international relations, warfare economics, and philosophy into a general theory of signal identification under observational ambiguity.

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Glossary

Bayesian Causal Inference Inference using probabilistic causal models and posterior updating.

Causal Constraint Manifold The set of realizations compatible with a causal structure.

Degeneracy Dimension The dimension of the realization manifold.

Entropy Collapse The transition from positive posterior entropy to zero entropy.

Identification Number The cardinality of the feasible generator set.

Inverse Statistical Geometry The study of constructing generators from prescribed statistical structures.

Observational Equivalence Distinct generators producing identical observables.

Realization Manifold The set of generators consistent with a fixed observable structure.

The End